# 校园实验室设备预约与借用管理系统

## 完整实体关系图谱

36个实体 / 289个字段 / 61条物理外键 / PostgreSQL

主键 · R01 → Parent.id · 外键与目标实体 · ? 可空 · UQ 单列唯一

实线：域内及主业务外键；共享责任主体引用以字段目标完整展开。

可编辑源文件可切换“共享引用”图层，检查全部61条连接。

## A 身份与权限

共享身份、岗位和管理范围

6 实体

### User A01

身份映射

core_user

| | | |
|---|---|---|
| id | bigint | |
| username | varchar(150) UQ | |
| school_no | varchar(32) UQ | |
| password | varchar(128) | |
| last_login | timestamptz ? | |
| is_superuser | boolean | |
| first_name | varchar(150) | |
| last_name | varchar(150) | |
| email | varchar(254) | |
| is_staff | boolean | |
| is_active | boolean | |
| date_joined | timestamptz | |
| display_name | varchar(60) | |

13 fields — 主键 / 外键 / 完整字段

### Role A02

业务角色

core_role

| | | |
|---|---|---|
| id | bigint | |
| code | varchar(20) UQ | |
| name | varchar(40) | |

3 fields — 主键 / 外键 / 完整字段

### LabGrant A04

实验室管理范围

core_labgrant

| | | |
|---|---|---|
| id | bigint | |
| user_id<br>R04 → User.id | bigint<br>A01 | |
| lab_id<br>R05 → Lab.id | bigint<br>B01 | |
| granted_at | timestamptz | |

4 fields — 主键 / 外键 / 完整字段

### UserRole A03

用户角色关联

core_userrole

| | | |
|---|---|---|
| id | bigint | |
| user_id<br>R01 → User.id | bigint<br>A01 | |
| role_id<br>R02 → Role.id | bigint<br>A02 | |
| granted_by_id<br>R03 → User.id | bigint ?<br>A01 | |
| granted_at | timestamptz | |

5 fields — 主键 / 外键 / 完整字段

### QualificationType A05

培训资格类型

core_qualificationtype

| | | |
|---|---|---|
| id | bigint | |
| code | varchar(32) UQ | |
| name | varchar(80) | |

3 fields — 主键 / 外键 / 完整字段

### UserQualification A06

用户培训资格

core_userqualification

| | | |
|---|---|---|
| id | bigint | |
| user_id<br>R06 → User.id | bigint<br>A01 | |
| qualification_id<br>R07 → QualificationType.id | bigint<br>A05 | |
| granted_by_id<br>R08 → User.id | bigint ?<br>A01 | |
| issued_at | timestamptz | |
| expires_at | timestamptz ? | |
| revoked | boolean | |

7 fields — 主键 / 外键 / 完整字段

## B 资源与开放

资源主档、类型明细和日历

7 实体

### Lab B01

实验室

core_lab

| | | |
|---|---|---|
| id | bigint | |
| code | varchar(32) UQ | |
| name | varchar(80) | |
| location | varchar(120) | |
| description | text | |
| is_active | boolean | |

6 fields — 主键 / 外键 / 完整字段

### OpeningPeriod B06

日间开放时段

core_openingperiod

| | | |
|---|---|---|
| id | bigint | |
| lab_id<br>R15 → Lab.id | bigint<br>B01 | |
| weekday | smallint | |
| starts_at | time | |
| ends_at | time | |
| valid_from | date | |
| valid_to | date ? | |

7 fields — 主键 / 外键 / 完整字段

### Accessory B05

设备配件档案

core_accessory

| | | |
|---|---|---|
| id | bigint | |
| resource_id<br>R14 → Resource.id | bigint<br>B02 | |
| code | varchar(32) | |
| name | varchar(80) | |
| quantity | integer | |
| is_active | boolean | |

6 fields — 主键 / 外键 / 完整字段

### CalendarException B07

单日开放例外版本

core_calendarexception

| | | |
|---|---|---|
| id | bigint | |
| lab_id<br>R58 → Lab.id | bigint<br>B01 | |
| actor_id<br>R59 → User.id | bigint<br>A01 | |
| day | date | |
| closed | boolean | |
| starts_at | time ? | |
| ends_at | time ? | |
| active | boolean | |
| reason | text | |
| created_at | timestamptz | |

10 fields — 主键 / 外键 / 完整字段

### Resource B02

可预约资源

core_resource

| | | |
|---|---|---|
| id | bigint | |
| lab_id<br>R09 → Lab.id | bigint<br>B01 | |
| code | varchar(40) UQ | |
| name | varchar(100) | |
| kind | varchar(16) | |
| description | text | |
| service_status | varchar(20) | |
| manual_hold | varchar(16) | |
| created_at | timestamptz | |
| updated_at | timestamptz | |

10 fields — 主键 / 外键 / 完整字段

### EquipmentDetail B03

设备属性

core_equipmentdetail

| | | |
|---|---|---|
| id | bigint | |
| resource_id<br>R10 → Resource.id | bigint UQ<br>B02 | |
| qualification_id<br>R11 → QualificationType.id | bigint ?<br>A05 | |
| access_room_id<br>R12 → Resource.id | bigint ?<br>B02 | |
| category | varchar(60) | |
| risk | varchar(20) | |
| physical_status | varchar(16) | |
| portable | boolean | |
| handover_location | varchar(120) | |

9 fields — 主键 / 外键 / 完整字段

### RoomDetail B04

场地属性

core_roomdetail

| | | |
|---|---|---|
| id | bigint | |
| resource_id<br>R13 → Resource.id | bigint UQ<br>B02 | |
| capacity | integer | |
| safety_note | text | |

4 fields — 主键 / 外键 / 完整字段

## C 预约与规则

申请、最终决定和规则版本

6 实体

### Reservation C01

资源预约

core_reservation

| | | |
|---|---|---|
| id | bigint | |
| applicant_id<br>R16 → User.id | bigint<br>A01 | |
| resource_id<br>R19 → Resource.id | bigint<br>B02 | |
| code | varchar(40) UQ | |
| starts_at | timestamptz | |
| ends_at | timestamptz | |
| purpose | text | |
| participants | integer | |
| status | varchar(16) | |
| approval_mode | varchar(10) | |
| request_key | uuid | |
| request_fingerprint | varchar(64) | |
| policy_snapshot | jsonb | |
| reason | text | |
| version | integer | |
| review_required | boolean | |
| review_reason | text | |
| arrived_at | timestamptz ? | |
| late_start_until | timestamptz ? | |
| created_at | timestamptz | |

20 fields — 主键 / 外键 / 完整字段

### ApprovalDecision C02

自动或人工确认决定

core_approvaldecision

| | | |
|---|---|---|
| id | bigint | |
| reservation_id<br>R22 → Reservation.id | bigint UQ<br>C01 | |
| actor_id<br>R23 → User.id | bigint ?<br>A01 | |
| mode | varchar(10) | |
| approved | boolean | |
| reason | text | |
| decided_at | timestamptz | |

7 fields — 主键 / 外键 / 完整字段

### ReservationReview C06

预约复核历史

core_reservationreview

| | | |
|---|---|---|
| id | bigint | |
| reservation_id<br>R54 → Reservation.id | bigint<br>C01 | |
| actor_id<br>R55 → User.id | bigint<br>A01 | |
| accepted | boolean | |
| reason | text | |
| snapshot | jsonb | |
| created_at | timestamptz | |

7 fields — 主键 / 外键 / 完整字段

### Policy C04

当前业务参数

core_policy

| | | |
|---|---|---|
| id | bigint | |
| version | integer | |
| window_days | integer | |
| slot_minutes | integer | |
| max_hours | integer | |
| equipment_quota | integer | |
| room_quota | integer | |
| simultaneous_equipment | integer | |
| grace_minutes | integer | |
| mock_delivery_failure | boolean | |
| updated_at | timestamptz | |

11 fields — 主键 / 外键 / 完整字段

### PolicyRevision C05

参数历史版本

core_policyrevision

| | | |
|---|---|---|
| id | bigint | |
| policy_id<br>R16 → Policy.id | bigint<br>C04 | |
| actor_id<br>R17 → User.id | bigint ?<br>A01 | |
| version | integer | |
| data | jsonb | |
| reason | text | |
| created_at | timestamptz | |

7 fields — 主键 / 外键 / 完整字段

### ReviewNote C03

审核补充记录

core_reviewnote

| | | |
|---|---|---|
| id | bigint | |
| reservation_id<br>R20 → Reservation.id | bigint<br>C01 | |
| actor_id<br>R21 → User.id | bigint<br>A01 | |
| kind | varchar(20) | |
| content | text | |
| evidence_ref | varchar(240) | |
| created_at | timestamptz | |

7 fields — 主键 / 外键 / 完整字段

## D 现场使用

设备交接与场地实际使用

5 实体

### Loan D01

设备借出事实

core_loan

| | | |
|---|---|---|
| id | bigint | |
| reservation_id<br>R24 → Reservation.id | bigint UQ<br>C01 | |
| operator_id<br>R25 → User.id | bigint<br>A01 | |
| code | varchar(40) UQ | |
| arrived_at | timestamptz | |
| service_started_at | timestamptz | |
| handed_over_at | timestamptz | |
| due_at | timestamptz | |
| tag_snapshot | varchar(40) | |
| condition_out | text | |

10 fields — 主键 / 外键 / 完整字段

### ReturnRecord D02

设备归还事实

core_returnrecord

| | | |
|---|---|---|
| id | bigint | |
| loan_id<br>R26 → Loan.id | bigint UQ<br>D01 | |
| operator_id<br>R27 → User.id | bigint<br>A01 | |
| code | varchar(40) UQ | |
| returned_at | timestamptz | |
| damaged | boolean | |
| missing | boolean | |
| fault | boolean | |
| note | text | |

9 fields — 主键 / 外键 / 完整字段

### RoomUse D04

场地实际使用与检查

core_roomuse

| | | |
|---|---|---|
| id | bigint | |
| reservation_id<br>R30 → Reservation.id | bigint UQ<br>C01 | |
| started_by_id<br>R31 → User.id | bigint<br>A01 | |
| ended_by_id<br>R32 → User.id | bigint ?<br>A01 | |
| arrived_at | timestamptz | |
| started_at | timestamptz | |
| actual_participants | integer | |
| start_note | text | |
| ended_at | timestamptz ? | |
| end_note | text | |
| abnormal | boolean | |

11 fields — 主键 / 外键 / 完整字段

### HandoverItem D03

配件交接快照

core_handoveritem

| | | |
|---|---|---|
| id | bigint | |
| loan_id<br>R28 → Loan.id | bigint<br>D01 | |
| accessory_id<br>R29 → Accessory.id | bigint ?<br>B05 | |
| code_snapshot | varchar(32) | |
| name_snapshot | varchar(80) | |
| expected_qty | integer | |
| out_qty | integer | |
| out_condition | varchar(160) | |
| return_qty | integer ? | |
| return_condition | varchar(160) | |

10 fields — 主键 / 外键 / 完整字段

### Coordination D05

预约协调凭据

core_coordination

| | | |
|---|---|---|
| id | bigint | |
| reservation_id<br>R56 → Reservation.id | bigint<br>C01 | |
| actor_id<br>R57 → User.id | bigint<br>A01 | |
| action | varchar(20) | |
| reason | text | |
| old_state | jsonb | |
| new_state | jsonb | |
| created_at | timestamptz | |

8 fields — 主键 / 外键 / 完整字段

## E 异常与维修

证据、工单和独立验收

5 实体

### Incident E01

资源异常事实

core_incident

| | | |
|---|---|---|
| id | bigint | |
| resource_id<br>R33 → Resource.id | bigint<br>B02 | |
| reporter_id<br>R34 → User.id | bigint<br>A01 | |
| reservation_id<br>R35 → Reservation.id | bigint ?<br>C01 | |
| confirmed_by_id<br>R36 → User.id | bigint ?<br>A01 | |
| code | varchar(40) UQ | |
| kind | varchar(20) | |
| description | text | |
| blocks_service | boolean | |
| occurred_at | timestamptz | |
| reported_at | timestamptz | |
| status | varchar(16) | |
| conclusion | text | |
| resolved_at | timestamptz ? | |

14 fields — 主键 / 外键 / 完整字段

### Evidence E02

异常证据索引

core_evidence

| | | |
|---|---|---|
| id | bigint | |
| incident_id<br>R37 → Incident.id | bigint<br>E01 | |
| uploaded_by_id<br>R38 → User.id | bigint<br>A01 | |
| file | varchar(100) | |
| original_name | varchar(160) | |
| sha256 | varchar(64) | |
| uploaded_at | timestamptz | |

7 fields — 主键 / 外键 / 完整字段

### IncidentResponse E05

异常确认与异议记录

core_incidentresponse

| | | |
|---|---|---|
| id | bigint | |
| incident_id<br>R60 → Incident.id | bigint<br>E01 | |
| actor_id<br>R61 → User.id | bigint<br>A01 | |
| kind | varchar(20) | |
| content | text | |
| created_at | timestamptz | |

6 fields — 主键 / 外键 / 完整字段

### WorkOrder E03

维修工单

core_workorder

| | | |
|---|---|---|
| id | bigint | |
| incident_id<br>R39 → Incident.id | bigint<br>E01 | |
| created_by_id<br>R40 → User.id | bigint<br>A01 | |
| assigned_to_id<br>R41 → User.id | bigint<br>A01 | |
| accepted_by_id<br>R42 → User.id | bigint ?<br>A01 | |
| code | varchar(40) UQ | |
| status | varchar(24) | |
| priority | varchar(12) | |
| result | text | |
| finished_at | timestamptz ? | |
| accepted_at | timestamptz ? | |
| created_at | timestamptz | |

12 fields — 主键 / 外键 / 完整字段

### WorkOrderEvent E04

维修及验收过程

core_workorderevent

| | | |
|---|---|---|
| id | bigint | |
| work_order_id<br>R43 → WorkOrder.id | bigint<br>E03 | |
| actor_id<br>R44 → User.id | bigint<br>A01 | |
| old_status | varchar(24) | |
| new_status | varchar(24) | |
| note | text | |
| created_at | timestamptz | |

7 fields — 主键 / 外键 / 完整字段

## F 审计与通知

状态区间和可追踪投递

7 实体

### AuditEvent F01

不可覆盖审计事件

core_auditevent

| | | |
|---|---|---|
| id | bigint | |
| actor_id<br>R45 → User.id | bigint ?<br>A01 | |
| lab_id<br>R46 → Lab.id | bigint ?<br>B01 | |
| dedupe_key | varchar(160) ? UQ | |
| actor_kind | varchar(8) | |
| entity_type | varchar(40) | |
| entity_id | bigint | |
| action | varchar(40) | |
| old_state | jsonb | |
| new_state | jsonb | |
| reason | text | |
| request_id | uuid | |
| created_at | timestamptz | |

13 fields — 主键 / 外键 / 完整字段

### ServicePeriod F02

资源服务状态区间

core_serviceperiod

| | | |
|---|---|---|
| id | bigint | |
| resource_id<br>R47 → Resource.id | bigint<br>B02 | |
| audit_id<br>R48 → AuditEvent.id | bigint UQ<br>F01 | |
| status | varchar(20) | |
| starts_at | timestamptz | |
| ends_at | timestamptz ? | |

6 fields — 主键 / 外键 / 完整字段

### MockReceipt F06

模拟投递去重凭据

core_mockreceipt

| | | |
|---|---|---|
| id | bigint | |
| delivery_id<br>R53 → Delivery.id | bigint UQ<br>F04 | |
| received_at | timestamptz | |

3 fields — 主键 / 外键 / 完整字段

### Notification F03

接收人通知

core_notification

| | | |
|---|---|---|
| id | bigint | |
| audit_id<br>R49 → AuditEvent.id | bigint<br>F01 | |
| recipient_id<br>R50 → User.id | bigint<br>A01 | |
| code | varchar(40) UQ | |
| kind | varchar(32) | |
| content | text | |
| created_at | timestamptz | |
| read_at | timestamptz ? | |

8 fields — 主键 / 外键 / 完整字段

### Delivery F04

渠道投递任务

core_delivery

| | | |
|---|---|---|
| id | bigint | |
| notification_id<br>R51 → Notification.id | bigint<br>F03 | |
| channel | varchar(16) | |
| status | varchar(16) | |
| retry_count | smallint | |
| next_attempt_at | timestamptz | |
| locked_until | timestamptz ? | |
| last_error | text | |

8 fields — 主键 / 外键 / 完整字段

### DeliveryAttempt F05

投递尝试

core_deliveryattempt

| | | |
|---|---|---|
| id | bigint | |
| delivery_id<br>R52 → Delivery.id | bigint<br>F04 | |
| attempt_no | smallint | |
| outcome | varchar(12) | |
| detail | text | |
| attempted_at | timestamptz | |

6 fields — 主键 / 外键 / 完整字段

### JobHeartbeat F07

后台任务健康记录

core_jobheartbeat

| | | |
|---|---|---|
| id | bigint | |
| name | varchar(40) UQ | |
| last_success | timestamptz ? | |
| last_error_at | timestamptz ? | |
| error_code | varchar(80) | |

5 fields — 主键 / 外键 / 完整字段

阅读顺序：资源 → 预约 → 现场使用 → 异常维修；身份、规则与审计为共用支撑。

实体代码用于跨域定位；R01-R61对应完整关系索引。实体、字段及外键由实际模型导出。

关键约束

待审核不占资源；确认与使用中的重叠区间互斥。

设备与场地分别记录实际使用；审计逻辑引用不伪装为外键。

2026-10-20 / ERD 2.1